

YASH VORA

Robotics & Embedded Systems Engineer

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SUMMARY

AI-driven mechatronics engineer with 2+ years on a commercial robotics product, developing embedded firmware and control software end to end — ESP32 firmware driving seven actuators, ESC-based BLDC motor control, piezoelectric sensing and BLE device networking — backed by an MS in Computer Science with an AI concentration. Work spans low-level motor control through to the mobile app and backend that drive the robot.

TECHNICAL SKILLS

Embedded & Firmware: C++, C, microcontrollers & SBCs (ESP32, MSP432, Raspberry Pi 4 & 5), PWM generation, ADC/GPIO, real-time control loops

Motor & Actuator Control: BLDC with ESCs, stepper drivers, servos, multi-axis coordination, ESC tuning & calibration

Hardware & Sensors: Piezoelectric sensor integration, ESC and servo interfacing, actuator wiring, soldering and robot assembly, hardware fault diagnosis and bench-level debugging

Robotics & Vision: ROS (coursework), OpenCV, 6-DOF arm control, machine vision classification, BLE device networks

Modeling & Simulation: MATLAB, Simulink, Simscape — battery, thermal and control-system modelling

Software: Python (FastAPI), React Native, TypeScript, JavaScript, SQL, Git, AWS

AI/ML: LLM integration, agentic workflows, prompt optimisation, speech-to-text, Scikit-Learn, NLTK

PROFESSIONAL EXPERIENCE

BodhiLabs (PongFox)

Bengaluru, Karnataka

AI & Full Stack Developer (Robotics)

Jul 2024 – Present

- Develop C++ firmware on ESP32 for a robotic table-tennis trainer driving seven actuators — 3× BLDC motors, 1× stepper and 3× servos — coordinating ball speed, spin, feed rate and 2-axis head aiming from a single control loop.
- Tune ESC parameters and PWM modulation across the BLDC drivetrain, holding shot-to-shot ball placement within ± 2.5 cm of target on a given unit despite mechanical variation between wheels.
- Engineered the sensor-integrated "SmartPad" accessory: conditioned piezoelectric sensor output into an ESP32 ADC to detect ball impacts and stream events over BLE to the robot controller, automating serve triggering.
- Reworked BLE pairing across robot, SmartPad and mobile app, replacing a scheme that required each device MAC address to be hard-coded into its counterpart with automatic discovery and connection — cutting per-unit pairing from 30–45 minutes of manual provisioning to roughly 5 seconds.
- Integrated an LLM-based voice pipeline converting natural speech into executable robot command sequences at 90% command accuracy, covering prompt design, response parsing and validation before commands reach the control firmware.
- Lead development of the cross-platform React Native app and Python/SQL backend behind it, delivering end-to-end latency of under 0.5 s from app command to motor response.
- Diagnose and resolve faults across the full stack — motor and ESC behaviour, sensor anomalies, BLE dropouts and firmware-to-app integration issues — isolating root causes between hardware, firmware and application layers.

General Engineering Corporation (GEC)

Pune, Maharashtra

Engineering Intern

Sep 2021 – Nov 2021

- Carried out the initial Raspberry Pi 4 integrations across all machines for an internal IoT system, bringing per-machine progress tracking into a single view.
- Analysed machining processes for cost and cycle-time efficiency, building Python data pipelines to prepare the underlying data.

PROJECTS

Battery Management System for Electric Vehicles — BTech Final Year Project

Jan 2022 – Jul 2022

- Modelled both active and passive cell-balancing BMS architectures in MATLAB/Simulink/Simscape, covering cell electrical behaviour, thermal effects, current measurement and state-of-charge estimation algorithms.
- Implemented monitoring protocols for charge/discharge cycling and temperature thresholds, with protective action on limit violation.
- Ran extended-duration and varied-load simulations to quantify capacity degradation prevented versus an unmanaged pack, and to validate system stability under load.

Robotics and AI for Object Sorting

May 2021 – Aug 2021

- Built a 6-DOF robotic arm sorting cell using OpenCV machine vision on Raspberry Pi to classify objects by visual features.
- Programmed pick-and-place control logic and calibrated the integrated system for real-time accuracy and repeatability.

EDUCATION

Pace University, Seidenberg School of CSIS

New York, NY | Sep 2022 – May 2024

Master of Science, Computer Science — Concentration: Artificial Intelligence

GPA: 3.82 / 4.0

Manipal Institute of Technology (MAHE)

Karnataka, India | Aug 2018 – Jul 2022

B.Tech, Mechatronics — Concentration: Electric Vehicle Technology

CGPA: 8.67 / 10

Relevant coursework: Robotics & ROS, Control Systems, Embedded Systems, Power Electronics, Machine Vision